CARE CONNECT

PROJECT REPORT

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CERTIFICATE

This is to certify that the project entitled "CARE CONNECT" is done and submitted by

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DECLARATION

I, **Mr. MUHAMMED RAMEEZ** here declare that the project entitled by **"CARE CONNECT"** submitted to the Calicut university in partial fulfillment of the requirements for the award of degree of **BACHELOR OF COMPUTER APPLICATION**, is a record of original work done by **HARIKRISHNAN K (IOAVBCA026), MUHAMMED SHIBIN FARHAN K (IOAVBCA016) ,YASEEN MALIK P (IOAVBCA043)** during their period of study at **ISS ARTS AND SCIENCE COLLEGE**, under the guidance of me.

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ABSTRACT

CARECONNECT

CareConnect is a compassionate app designed to bridge the gap between orphanages in need and caring individuals looking to make a positive impact. With two distinct modules – Orphanage and Users – the app facilitates seamless communication and support. Orphanages can submit requests for vital needs like food, clothing, education, and fundraising, while also maintaining a comprehensive list of orphaned children. Users, on the other hand, can browse orphanage profiles, respond to specific requests, access detailed information about the children in need, and even initiate adoption requests. CareConnect acts as a digital conduit, fostering connections and channeling support to create a brighter future for orphaned children.

Modules:

1. Orphanage Module:

-Login, Orphanage profile management.

-Uploading orphans list and details.

-View user profile.

-Sending requests to users for vital needs like food, clothing, education, and fundraising.

-Viewing replies and adoption requests from users.

2. User Module:

-Login, User profile management.

-Explore orphanages.

-View orphanage profile.

-View orphans lists and details.

-View requests from orphanages and reply

-Request adoption of an orphan

CONTENTS

CHAPTERS **PAGE**

1. INTRODUCTION……..1

1.1 About the organization……2

1.2 About the System……2

2. SYSTEM ANALYSIS……3

2.1 Abstract……..4

2.2 Existing system……..4

2.2 Proposed system……4

3. FEASIBILITY STUDY……..5

3.1 Technical feasibility……...6

3.2 Economical feasibility……7

3.3 Operational feasibility……...7

3.4 Behavioral feasibility……7

4. SOFTWARE ENGINEERING PARADIGM……..8

4.1 Agile Model……...9

4.2 Scrum……...12

4.3 SYSTEM REQUIRMENTS SPECIFICATION…….13

4.3.1Frontend…….....15

4.3.2 Flutter…….....15

4.4 Backend…….....17

4.3.1 Firebase…….....17

5. SYSTEM DESIGN…….....18

5.1 Input Design…….19

5.2 Output Design……..21

5.3 Normalization……..22

5.4 Modules & Description……...23

5.5 Data flow diagram……...24

5.6 Database Design……...31

5.7 Database Table…….32

6.SYSTEM TESTING & IMPLEMENTATION……...40

6.1 System testing……...41

6.2 System implementation……41

7. SYSTEM MAINTAINANCE……...46

7.1 Corrective Maintenance……...47

7.2 Adaptive Maintenance…….....47

7.3 Perfective Maintenance……47

7.4 Preventive Maintenance……...47

8. FUTURE ENHANCEMENT……48

9.CONCLUSION……....51

10.BIBLIOGRAPHY……..53

11.APPENDIX……55

11.1 Samplescreenshot……56

11.2 Sample codes……...…..57

INTRODUCTION

1. About the system

Careconnect

CareConnect stands as a compassionate application, strategically designed to establish meaningful connections between orphanages and individuals with a heart for positive impact. Operating through two distinct modules, the app seamlessly facilitates communication and support. The Orphanage module enables institutions to submit requests for essential needs such as food, clothing, education, and fundraising. Additionally, it serves as a repository for comprehensive profiles of orphaned children, ensuring a holistic approach to their well-being.

On the flip side, the Users module empowers individuals to actively engage with orphanages, responding to specific needs, accessing detailed information about children in need, and even initiating adoption requests. This user-friendly interface transforms CareConnect into more than just a communication platform it's a conduit for heartfelt connections and tangible support. By bridging the gap between orphanages seeking assistance and caring individuals ready to make a difference, the app becomes a catalyst for positive change, embodying the spirit of compassion and community.

CareConnect, therefore, transcends the conventional boundaries of digital applications. It acts as a dynamic bridge, fostering connections that lead to a brighter future for orphaned children. Through its innovative approach, the app not only streamlines communication but also ensures that the support extended is targeted, making a lasting impact on the lives of those in need. In essence, CareConnect is not just an app; it's a compassionate ecosystem where generosity meets necessity, creating a harmonious synergy that echoes the shared goal of building a better tomorrow for orphaned children.

SYSTEM ANALYSIS

2. SYSTEM ANALYSIS

System study is a process of studying a procedure to identify the objectives and purposes of a system as well as to analyze the existing system's problems and drawbacks. The system analysis will create systems and procedures that will achieve these objectives as well as solutions to the problems in an efficient way. It is the process of gathering and interpreting facts, diagnosing problems and using the information to recommended improvements to the system. Training, experience and common sense are required for the collection of the information needed to do the analysis.

2.2 Existing system

There is currently no well-defined system exclusively dedicated to orphans. Existing systems lack direct mechanisms to connect individuals with orphanages for comprehensive assistance, and they also lack adoption request facilities. Additionally, there is a notable absence of an advanced database to manage the data of orphaned children.

2.3 Proposed system

The envisioned system seeks to establish a well-defined platform exclusively tailored for orphans. It aims to address existing shortcomings by implementing direct mechanisms to connect individuals with orphanages for comprehensive assistance. The proposed system will incorporate adoption request facilities to facilitate the adoption process seamlessly. Furthermore, it will integrate an advanced database system to efficiently manage and store data related to orphaned children, enhancing overall transparency and accessibility.

FEASIBILITY STUDY

3. FEASIBILITY STUDY

A feasibility study is a preliminary study undertaken to determine and document a project's viability.
The results of this study are used to decide whether to proceed with the project.
It is an analysis of possible alternative solutions to a problem and recommendation on the best alternative. The feasibility study proposes one or more conceptual solutions to the problem set for the project. The conceptual solution gives an idea of what the new system will look like. The feasibility study environment enables all alternatives to be discussed and evaluated. This phase starts with an identification of the main characteristics of the required system. During this stage it is important to collect as much information as possible about the software package that might meet the specification from as many sources as possible. Normally, feasibility study is a cost benefit analysis of various alternatives. It can be defined as a systematic comparison between the cost of carrying out a service or activity and the value of that service or activity. The main benefits are qualitative rather than quantitative. A feasibility study could be used to test a new working system, which could be used because:

- The current stem may no longer suit its purpose
- Technological advancement may have rendered the current system obsolete
- The business is expanding allowing it to cope with extra workload
- Customers are complaining about the speed and quality of work the business provides

Feasibility study is carried out to determine whether the proposed system is possible to develop with available resources and what should be the cost consideration. Facts considered in the feasibility analysis are:

- Technical Feasibility
- Economic Feasibility
- Operational Feasibility
- Behavioral Feasibility

3.1 Technical Feasibility

Technical Feasibility study is performed to check whether the proposed system is technically Feasible or not. Technical feasibility centers on the existing computer system (hardware, software, etc.) and to what extent it can support the proposed addition. This involves financial consideration to accommodate technical enhancement. This system is technically feasible. All the data is stored in files. The input can be done through dialog boxes which are both interactive and user friendly. Hard copies can be obtained for future use, by diverting the documents to a printer. Windows serves as the platform for the new system. This project was developed using the Flutter framework which is technically feasible as it requires only the existing technology.

3.2 Economical Feasibility

Economic Feasibility Study is the most frequently used method for evaluating the effectiveness of a candidate system. More commonly known as cost/benefit analysis, the procedure is to determine the benefits and savings that are expected from a candidate system and compare them with cost. This analysis phase determines how much cost is needed to produce the proposed system. This system is economically feasible since it does not require any initial setup. It requires only network connection and the cost needed to host the site.

3.3 Operational Feasibility

Operational Feasibility study is performed to check whether the system is operationally Feasible or not. Using command buttons throughout the application programs enhances operational feasibility. So, maintenance and modification is found to be easier. The purpose of this feasibility is to determine whether the new system will be used if it is developed and implemented. Usually, people are reluctant to change their progression. Hence an additional Effort is to be made to train and educate the users on the new way of the system. In this project the system is well coded and designed such that when it is compiled it is well developed and installed. We can do maintenance and modifications easily. There is no difficulty in implementing the system and the proposed system is user friendly and functionally reliable and thus it is operational feasible.

3.4 Behavioral Feasibility

This analysis involves how it will work when it is installed and the assessment of the political and managerial environment in which it is implemented. People are inherently resistant to change and computers have been known to facilitate change. An estimate should be made about the reaction of the user staff towards the development of the computerized system. The introduction of a candidate system requires special effort to educate, sell and train the staff for conducting the business. This includes the following questions:

- Is there sufficient support for the users?
- Will the proposed system cause harm?
- This system is behavior feasible since the system sufficiently supports the users and the system does not cause any harm to the users.

The project would be beneficial because it satisfies the objectives when developed and installed. All the behavioral aspects are considered carefully and conclude that the project is behaviorally feasible.

SOFTWARE ENGINEERING PARADIGM

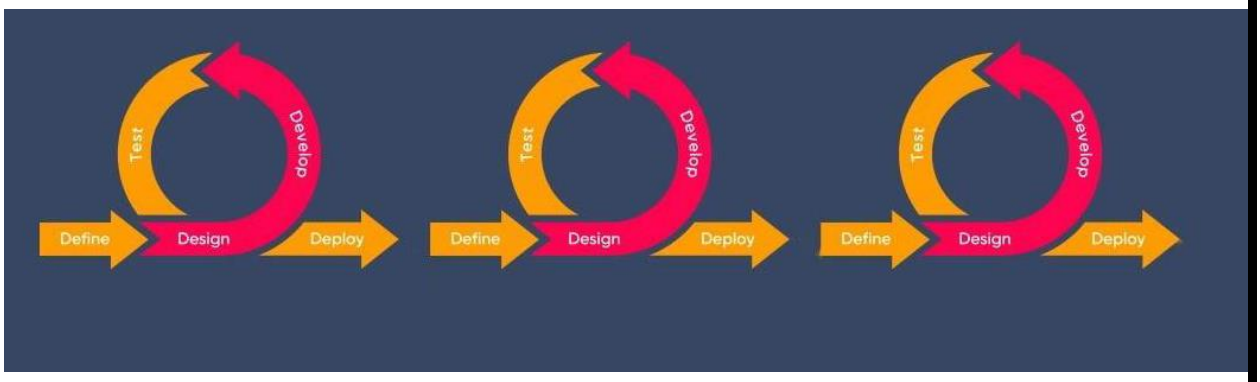
4. SOFTWARE ENGINEERING PARADIGM

The software engineering paradigm which is also referred to as a software process model or software development life cycle (SDLC model is the development strategy that encompasses the process, methods and tools. SDLC describes the period of that time starts with the software system being conceptualized.

4.1 AGILE MODEL

Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software products. Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. Every iteration involves cross functional teams working simultaneously on various areas. The Agile model believes that every project needs to be handled differently and the existing methods need to be tailored to best suit the project requirements. In Agile, the tasks are divided into time boxes (small time frames) to deliver specific features for a release. Iterative approach is taken and working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer.

Graphical illustration of the Agile Model :



Here are the important stages involved in the Agile Model process in the SDLC life cycle:

Requirements gathering and analysis

In this phase, you must define the requirements. You should explain business opportunities and plan the time and effort needed to build the project. Based on this information, you can evaluate technical and economic feasibility.

Design the requirements

When you have identified the project, work with stakeholders to define requirements. You can use the user flow diagram or the high-level UML diagram to show the work of new features and show how it will apply to your existing system.

Construction/ iteration

When the team defines the requirements, the work begins. Designers and developers start working on their project, which aims to deploy a working product. The product will undergo various stages of improvement, so it includes simple, minimal functionality.

Testing

In this phase, the Quality Assurance team examines the product's performance and looks for the bug.
Deployment
In this phase, the team issues a product for the user's work environment. Feedback After releasing the product, the last step is feedback. In this, the team receives feedback about the product and works through the feedback.

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Feedback

After releasing the product, the last step is feedback. In this, the team receives feedback about the product and works through the feedback.

Advantages of Agile Model

□ Is a very realistic approach to software development.

□ Promotes teamwork and cross training.

□ Functionality can be developed rapidly and demonstrated.

□ Resource requirements are minimum.

□ Suitable for fixed or changing requirements.

□ Delivers early partial working solutions.

□ Good model for environments that change steadily.

□ Minimal rules, documented.

□ Enables concurrent development and delivery within an overall planned context.

□ Little or no planning required.

□ Easy to manage.

Disadvantages of Agile Model

□ Not suitable for handling complex dependencies.

□ More risk of sustainability, maintainability and extensibility.

□ An overall plan, an agile leader and agile PM practice is a must without which it will not work.

□ Strict delivery management dictates the scope, functionality to be delivered, and adjustments to meet the deadlines.

□ Depends heavily on customer interaction, so if customer is not clear, team can be driven in the wrong direction.

4.1 Scrum

Scrum is an agile process most commonly used for product development. especially software development. Scrum is a project management framework that is applicable to any project with aggressive deadlines, complex requirements and a degree of uniqueness. In Scrum, projects move forward via a series of iterations called sprints. Each sprint is typically two to four weeks long. An introduction to Scrum would not be complete without knowing the Scrum terms you'll be using. This section in the Scrum overview will discuss common concepts in Scrum.

Scrum team: A typical scrum team has between five and nine people, but Scrum projects can easily scale into the hundreds. However, Scrum can easily be used by one person teams and often is. This team does not include any of the traditional software engineering roles such as programmer, designer, tester or architect. Everyone on the project works together to complete the set of work they have collectively committed to complete within a sprint. Scrum teams develop a deep form of camaraderie and a feeling that "we're all in this together.

SYSTEM REQUIREMENTS SPECIFICATION

4. SYSTEM REQUIREMENTS

4.1 Hardware Requirements

The selection of hardware is very important in the existence and proper working of any software. Then selection hardware, the size and capacity requirements are also important.

• Processor : Intel i5

• Primary Memory : 4GB RAM and above

• Storage : 20 GB hard disk and above

• Display : Generic PnP Monitor

• Keyboard : Windows compatible

• Mouse : Standard Mouse

4.1 Software Requirements

One of the most difficult task is selecting software for the system, once the system requirements is found out then we have to determine whether a particular software package fits for those system requirements. The application requirement:

OPERATING SYSTEM: Windows 10 for better performance

FRONT END: Dart BACK END: Firebase

SOFTWARE USED: Visual Studio Code

FRAME WORK: Flutter

4.1 Frontend

4.1 Flutter

Flutter is an open-source mobile application development framework created by Google. It was first introduced in 2017 and has gained popularity among mobile developers due to its ease of use and efficiency in building cross platform applications for both Android and iOS devices. Flutter uses a programming language called Dart, which is also developed by Google, and has its own set of widgets and libraries that enable developers to build beautiful and responsive user interfaces. Flutter's hot reload feature allows developers to see the changes they make to their code in real-time, making the development process faster and more efficient. One of the key advantages of Flutter is that it provides a single codebase for both Android and iOS platforms, reducing the time and effort needed to develop and maintain separate applications for each platform. Additionally, Flutter apps can offer high performance and smooth animations, which can enhance the user experience.Flutter also provides a range of features and tools for developers, such as debugging, testing, and integration with other development tools.

4.1.1 Dart

Dart is a client-optimized programming language for apps on multiple platforms. It is developed by Google and is used to build mobile, desktop, server, and web applications Dart is an object-oriented, class-based, garbage collected language with C-style syntax.

Dart can compile to either native code or JavaScript. It supports interfaces, mixins, abstract classes, reified generics, and type inference.

There are four ways to run Dart code:

4.1.2 Compiled as JavaScript

To run in mainstream web browsers, Dart relies on a source-to-source compiler to JavaScript. According to the project site, Dart was designed to be easy to write development tools for, well-suited to modern app development, and capable of high performance implementations. When running Dart code in a web browser the code is precompiled into JavaScript using the dart js compiler. Compiled as JavaScript, Dart code is compatible with all major browsers with no need for browsers to adopt Dart. Through optimizing the compiled JavaScript output to avoid expensive checks and operations, code written in Dart can, in some cases, run faster than equivalent code hand-written using JavaScript idioms.

4.1.3 Stand-alone

The Dart software development kit (SDK) ships with a stand-alone Dart VM, allowing Dart code to run in a command-line interface environment. As the language tools included in the Dart SDK are written mostly in Dart, the standalone Dart VM is a critical part of the SDK. These tools include the dart2js compiler and a package manager called pub. Dart with a complete standard library allowing users to write fully working system ships apps, such as custom web servers.

4.1.4 Ahead-of-time compiled

Dart code can be AOT-compiled into machine code (native instruction sets). Apps built with Flutter, a mobile app SDK built with Dart, are deployed to app stores as AOT- compiled Dart code.

4.1.5 Native

Dart 2.6 with dart native compiler to compile to self-contained, native executable code. Before Dart 2.6, this feature only exposed this capability on iOS and Android mobile devices via Flutter.

4.1.6 Visual Studio Code

Visual Studio Code (VS Code) is a lightweight yet powerful source code editor developed by Microsoft. With its sleek and intuitive interface, VS Code provides a seamless development experience across various programming languages. The editor supports syntax highlighting, code completion, and intelligent suggestions, enhancing productivity for developers. Its extensibility is a standout feature, allowing users to customize their environment with a wide range of extensions available through the Visual Studio Code Marketplace. VS Code also integrates seamlessly with version control systems like Git, making it a preferred choice for collaborative coding projects. Its versatility and efficiency, combined with features like debugging, integrated terminal, and a rich set of extensions, make Visual Studio Code a popular choice among developers for creating, editing, and debugging code across different platforms

4.2 Back End

4.2.1 Firebase

Firebase is a cloud-based platform provided by Google that offers a variety of services to develop and operate web and mobile applications. One of the core services provided by Firebase is the Firebase Realtime Database, which is a NoSQL cloud-hosted database that allows real-time synchronization of data across multiple clients. Firebase is designed to handle complex and large-scale data and can be used to store various types of data including strings, numbers, Boolean values, and nested objects. With its real-time synchronization capabilities, data stored in the Firebase Realtime Database can be updated in real-time across all clients connected to the database, which can help reduce data inconsistencies. Firebase can be an effective tool for storing and managing data in a structured way for web and mobile applications, and documenting its usage can help ensure that the database is being used effectively and efficiently in a project.

Features of Firebase

• Automatically synchronizes data across all connected clients in real-time.

• Database can be accessed and updated even when the device is offline

• Designed to handle large applications with thousands of concurrent users.

• Allows flexible data storage and retrieval without predefined schema.

SYSTEM DESIGN

5. SYSTEM DESIGN

Design is a multi-step process that focuses on data structure, software architecture, procedural details and interface between the modules. The design process also translates the requirements into the representation of the software that can access for quality before coding begins. Design is the only way that can accurately translate a customer's finished software product. The most creative and challenging phase of the system development is the system design. It provides the understanding and procedural details necessary for implementing the system recommended in the feasibility study. Design goes through the logical and physical stages of development. System design goes through two phases of development:

• Logical design

• Physical design

Logical Design

The part of the design process that is independent of any specific hardware or software platform is referred to as logical design. During logical design, all functional features of the system chosen for development in the analysis phase are described independently of any computer platform. Logical design concentrates on the business aspects of the system and tends to be oriented to a high level of specificity. During the logical design of the proposed system, all the functional features of the system which are identified during the analysis phase are collected and properly specified independent of any hardware and software specifications.

Physical Design

During physical design the logical specifications of the system from logical design are transferred into technology specific details from which all programming and system construction can be accomplished.

5.1 Input Design

In input design, user originated inputs are converted into computer based format. In output design, the emphasis is on producing the hard copy of the information requested or displaying the output on the screen in a predefined format. The following features have been incorporated into the input design of the proposed system.

Easy Data Input

Appropriate messages are provided in the message area, which prompts the user to enter the right data. Erroneous data inputs are checked at the end of each screen entry.

Data Validation

The input data is validated to minimize errors in data entry. For certain, data specific codes have been given and validation is done which enables the user to enter the required data or correct them if they entered wrong inputs.

User Friendliness

User is never left in a state of confusion as to what is happening, instead appropriate errors and acknowledge messages are sent. Bootstrap's alert messages are being used to indicate the error, warning messages.

Consistent Format

A fixed format is adopted for displaying the alert messages. Every screen has a line, which displays the operation that can be performed after the data entry. They are normally done at the touch of a key.

Interactive Dialogue

The system engages the user in an interactive dialogue. The system is able to extract missing or omitted information from the user by directing the user through appropriate messages, which are displayed.

5.2 Output Design

Output design generally refers to the results and information data generated by the system. For many end users, output is the main reason for developing the system and the basis on which they evaluate the usefulness of application. The objective of a system finds its shape in terms of the output. The analysis of the objective of a system leads to determination of outputs. Outputs of a system can take various forms. The most common are reports, screen displays, printed form, graphical drawings etc. The output also varies in terms of their contents, frequency, timing and format. The output from a system is the justification for its existence. If the outputs are inadequate in any way, the system itself is inadequate. The basic requirement of output is that it should be accurate, timely and appropriate in terms of content, medium and layout for its intended purpose. Hence it is necessary to design output, so that the objective of the system is met in the best possible manner. When designing output, the system analyst must accomplish things like, to determine what information to be present, to decide whether to display or print the information and select the output medium to distribute the output to intended recipients. External outputs are those, whose destination will be outside the organization and which requires special attention as the project image of the organization. Internal output is those whose destination within the organization. It is to be carefully designed, as they are the user's main interface with the system. Interactive outputs are those which the user uses to communicate directly with the computer.

5.3 Normalization

Normalization is the process of decomposing a set of relations with anomalies to produce smaller and well-structured relations that contain minimum redundancy. It is a formal process of deciding which attributes should be grouped together in a relation. The purpose of normalization is to make table as simple as possible. Normalization carried out in the system for following reasons.

• To structure the data. So that there is no repetition of data, this helps in saving space.

• To permit simple retrieval of data in response to query and request.

• To simplify maintenance of data through updates, insertion and deletion.

• To reduce the need to restructures or recognize data which new application requirements arise.

Primary key is assigned for this purpose. The primary key fields in almost all tables help to ease the search and improve efficiency.

The different normal forms applied during the database design are given below.

> First Normal Form (INF)

>Second Normal Form (2NF)

> Third Normal Form (3NF)

FIRST NORMAL FORM:

First normal form decomposes all data into two dimensional records. First normal form is achieved when all repeating groups are removed so that a record is of fixed length.

SECOND NORMAL FORM:

Second normal form eliminates any relationships in which data elements do not fully dependent on the primary key of the record. Second normal form is achieved when a record is of INF and each item in the record is fully functionally depending on the primary key for identification.

THIRD NORMAL FORM:

Third normal form eliminates any relationships that contain transitive Xa hold in R and is not in X, then either X is super key for R or A is prime (1.e. A is a dependencies. A relational R is said to be in 3NF, if whenever a functional dependencies, subset of the candidate key). In our project the coding are done using python. This language is more powerful.

5.4 Modules & Description

1. Orphanage Module:

-Login, Orphanage profile management.

-Uploading orphans list and details.

-View user profile.

-Sending requests to users for vital needs like food, clothing, education, and fundraising.

-Viewing replies and adoption requests from users.

2. User Module:

-Login, User profile management.

-Explore orphanages.

-View orphanage profile.

-View orphans lists and details.

-View requests from orphanages and reply

-Request adoption of an orphan

5.5 Data flow diagram

The database may be defined as an organized collection of related information. The organized information serves as a base from which further recognition can be retrieved desired information or processing the data. The most important aspect of building an application system is the design of tables. The data flow diagram is used for classifying system requirements for major transformation that will become programs in system design. This is the starting point of the design phase that functionally decomposes the required specifications down to the lower level of details. It consists of a series of bubbles joined together by lines.

• Bubbles: Represent the data transformations.

• Lines: Represents the logic flow of data.

Data can trigger events and can be processed to useful information. System analysis recognizes the central goal of data in organizations. This dataflow analysis tells a great deal about how organization objectives are accomplished. Dataflow analysis studies the use of data in each activity. It documents this finding in DFD's. Dataflow analysis gives the activities of a system from the viewpoint of data where it originates, how they are used or hanged or where they go, including the stops along the way from their destination. The components of dataflow strategy span both requirements determination and system's design. The first part is called dataflow analysis. As the name suggests, we didn't use the dataflow analysis tools exclusively for the analysis stage but also in the designing phase with documentation.

Constructing a DFD

Several rules of thumb are used in drawing DFDs: -

• Processes should be named and numbered for easy reference.

• The direction of flow is from source to destination, although they may flow back to a source. One way to indicate this is to draw a long flow line back to the source. An alternative way is to repeat the source symbol as a destination.

• When a process is exploded into lower-level details, they are numbered.

• A level 0 DFD, also called a context level, represents the entire software elements as a single bible with input and output indicated by incoming and outgoing arrows respectively. Additional process and information flow parts are represented in the next level i.e. Level 1 DFD. Any process, which is complex in Level 1, will be further represented into sub functions in the next level, i.e. Level 2 DFD is a means of representing a system at any level of detail with a graphic network of symbols showing data flows, data stores, data process, sources or destination. The DFD is designed to aid communication. DFD shows the minimum contents of data stores. In

order to show what happens within a given process, then the detailed explosion of that process is shown. The DFD methodology is quite effective, especially when the required design is unclear and the user and the analyst need a notational language for communication.

Context Diagram

The top-level diagram is often called a "context diagram". It contains a single process, but it plays a very important role in studying the current system. The context diagram defines the system that will be studied in the sense that it determines the boundaries. Data flow Diagrams are made up of a number of symbols which represent system components. Data flow modelling uses four kinds of symbols.

Process

Process shows work of the system. Each process has one or more data inputs and produces one or more data outputs. Processes are represented by circles in DFDs.

Data Stores

Data stores are the repositories of data. Processes can enter data into the store or retrieve the data from the data store. Data stores are represented by parallel lines which may be depicted horizontally or vertically.

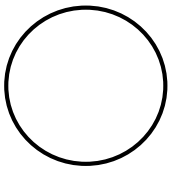
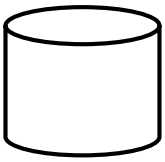
Data Flows

The arrows represent data flow. A data flow is data in motion. Data flow represents an input of data to a process or the output of the data from a process.

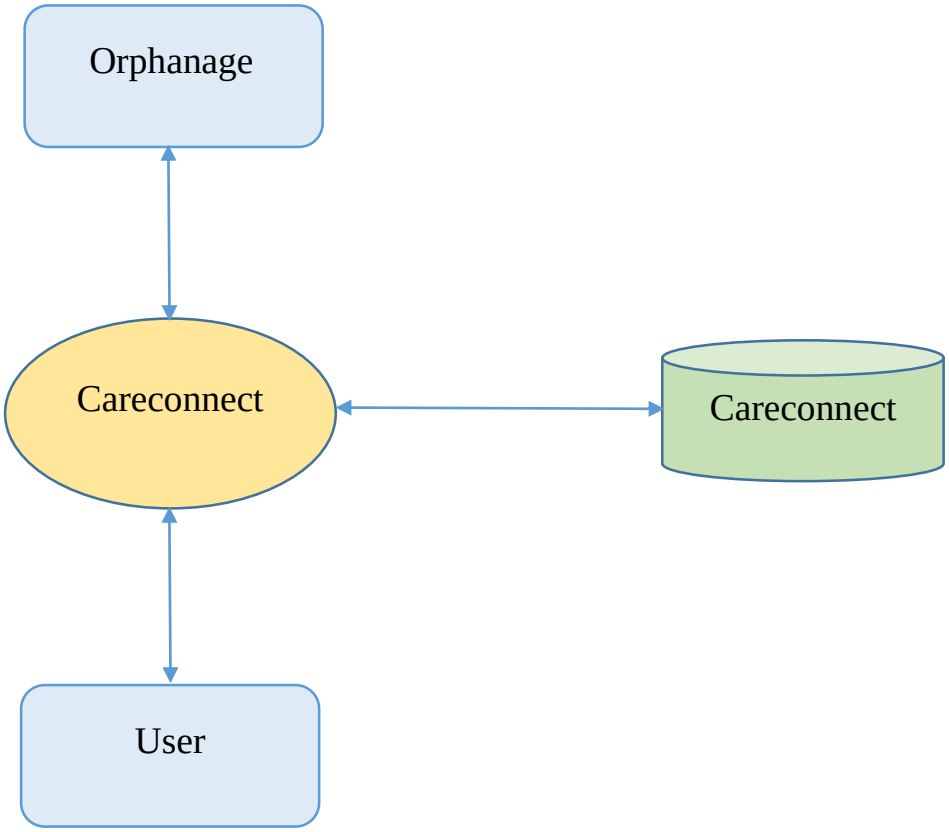
External Entities

External entities are outside the system but they either supply input to the system or use other system outputs. They are entities on which the designer has control. External entities that supply data into the system are sometimes called source. External entities that use the system are called sinks, represented by rectangles.

Data Flow Diagram Symbols

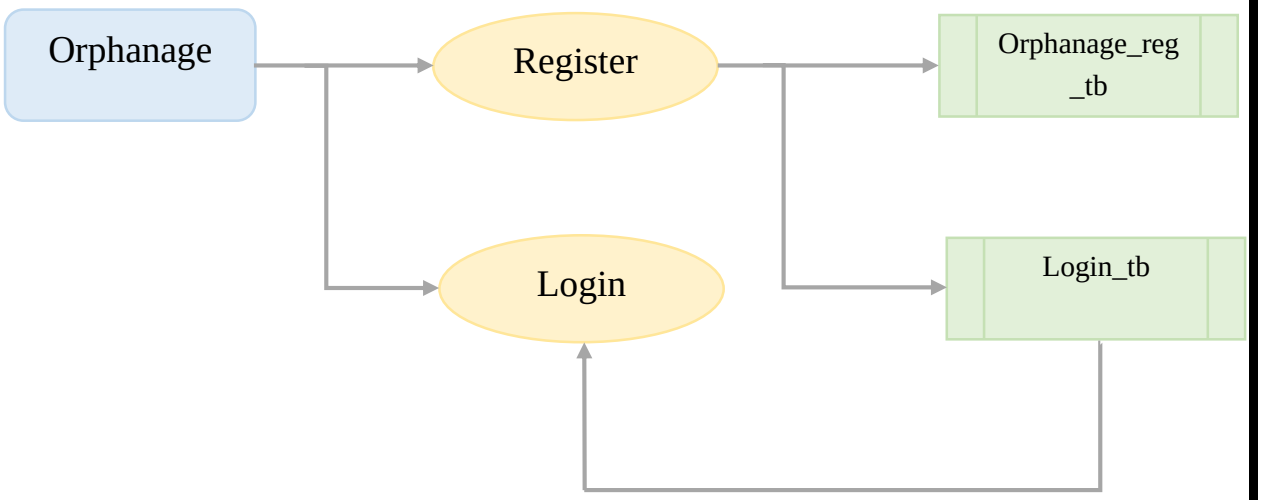
Symbol	Name	Function
	**Process**	Performs Some transformation of input data to yield output data.
	Data flow	Used to connect processes to each other, sources or sinks, the arrow head indicates direction of data flow.
	Source or sink (External Entity)	A source of inputs or sinks of system outputs
	**Data Store** (Internal Storage)	A repository of data

Level 0

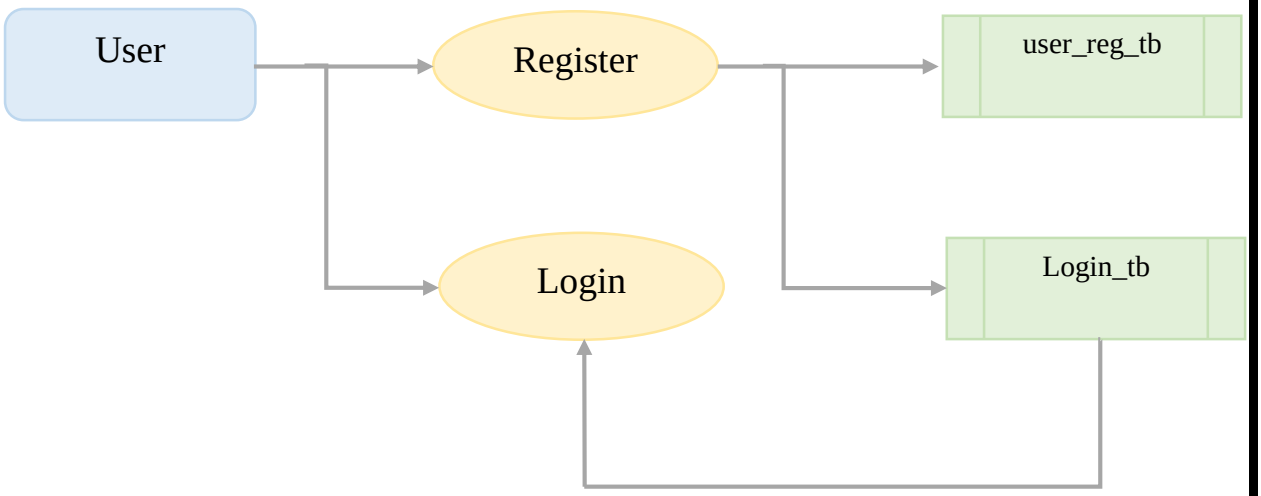


Level 1

1.1

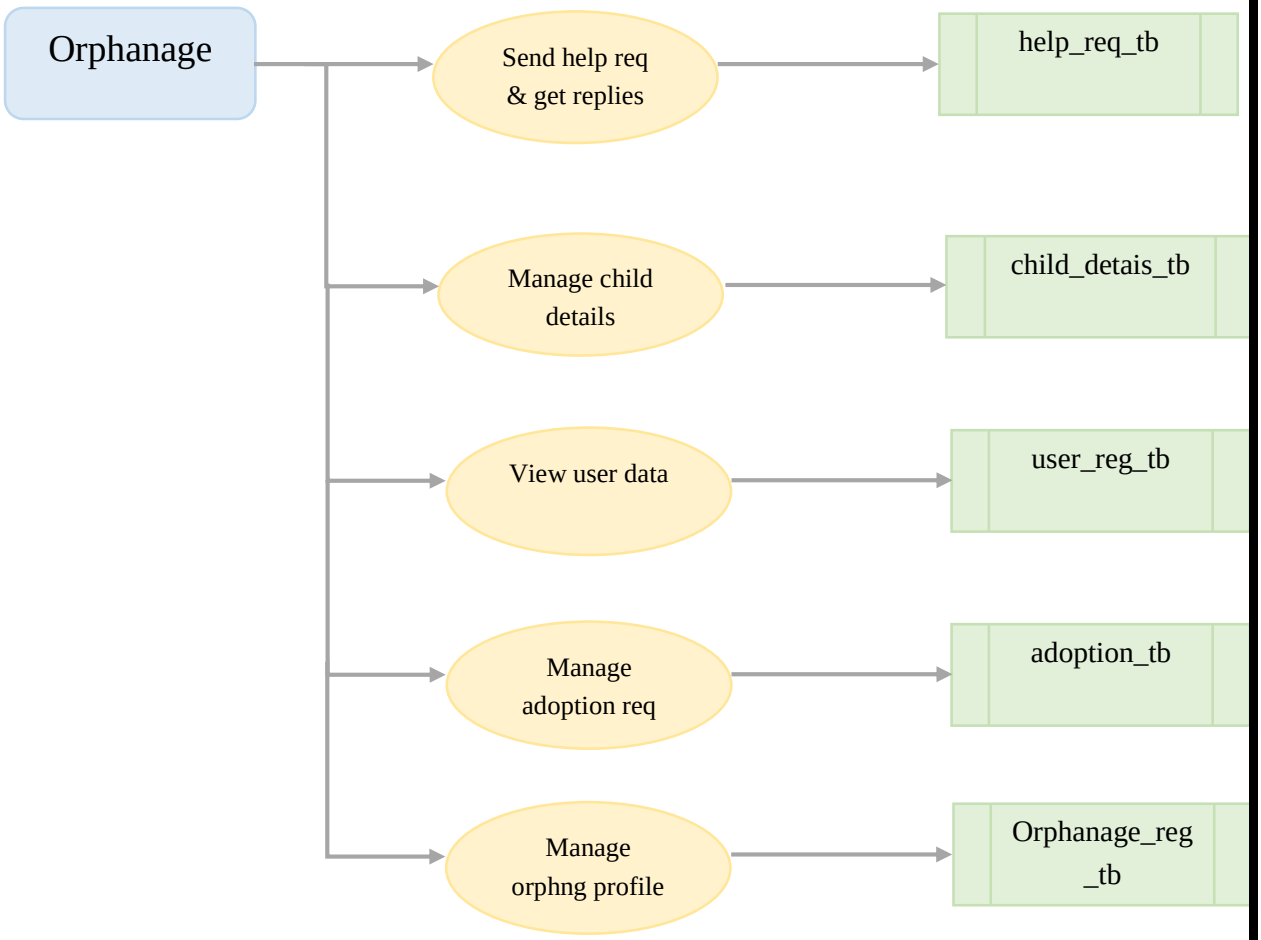

Orphanage
Register
Login
Orphanage_reg
_tb
Login_tb

1.2

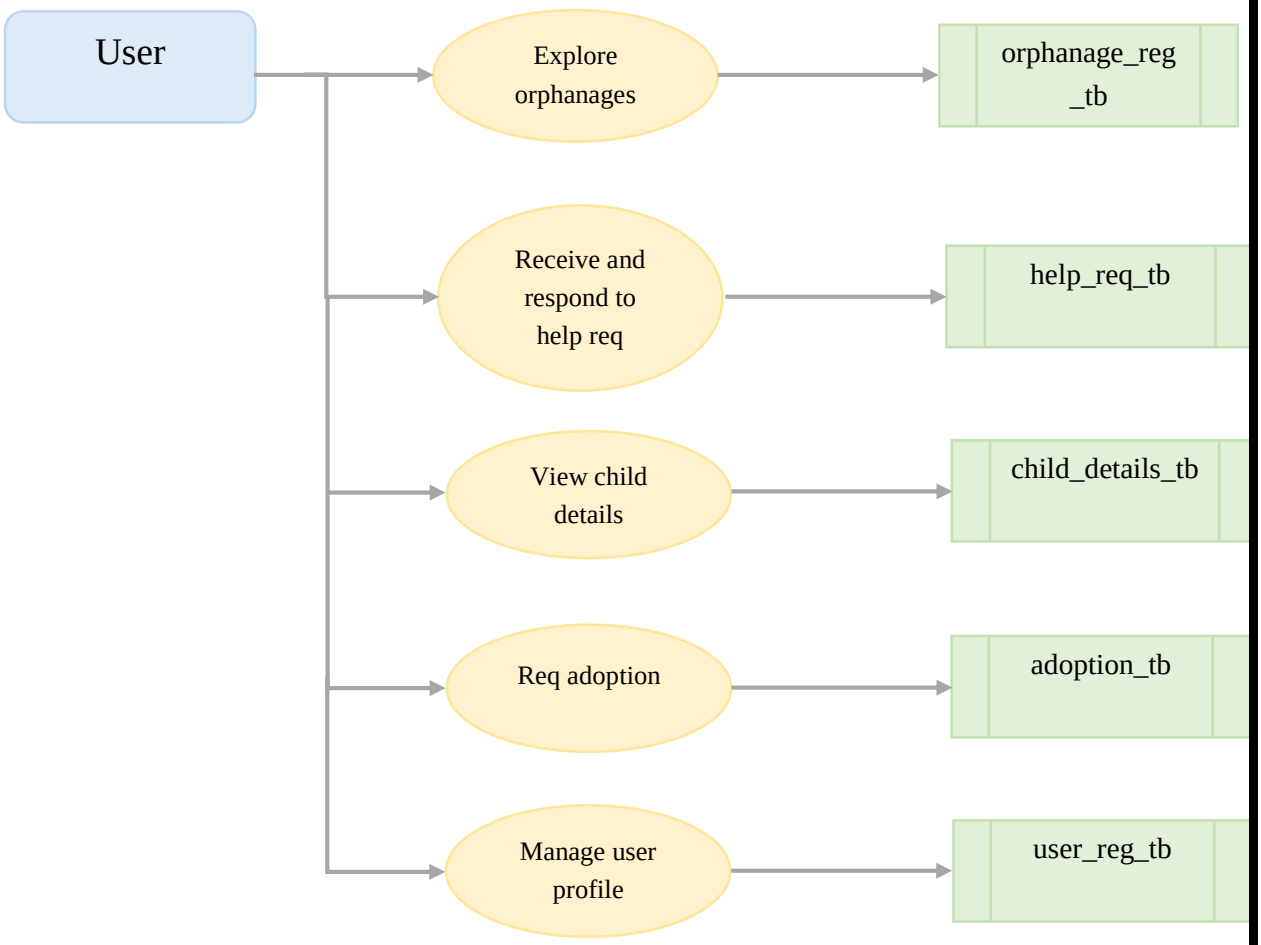

User
Register
Login
user_reg_tb
Login_tb

Level 2

2.1


Orphanage
Send help req & get replies
help_req_tb
Manage child details
child_detais_tb
View user data
user_reg_tb
Manage adoption req
adoption_tb
Manage orphng profile
Orphanage_reg _tb

2.2


User
Explore orphanages
orphanage_reg _tb
Receive and respond to help req
help_req_tb
View child details
child_details_tb
Req adoption
adoption_tb
Manage user profile
user_reg_tb

5.6 Database Design

A data base is a collection of inter related data stored with minimum redundancy to serve many quickly and efficiently. The general way is to make information accessing easy, quick, inexpensive and flexible for the user. In data base design several objectives are considered controlling redundancy, ease of learning and use data dependence, more information at low cost, accuracy and integrity are some of them. The design should be done in a way the information stored in the database can retrieved quickly whenever necessary. The general theme behind a database is to handle information as an interfered whole. A database is a collection of interrelated data stored with minimum redundancy to serve users quickly and efficiently. Database design runs parallel without application design. As we collect information about what is to be done, we will obviously collect information about data need to entered, stored messages and printed reports. The designing of database is done outmost care and security during the designing phase of the system. Special care was taken to develop minimum number of databases for the maximum efficiency of the system.

The organization of the database aims to achieve the following objectives,

- Data independence
- Recovery from failures
- Privacy and security
- Performance
- Easy to learn in use

Login Table

FIELD	TYPE	CONSTRAINTS
login_id	int	primary key
e- mail	varchar	not null
password	varchar	not null
type	varchar	not null

Orphanage registration Table

FIELD	TYPE	CONSTRAINTS
Orphanage_id	int	primary key
Orphanage_name	varchar	not null
about	varchar	not null
no of children	int	not null
Contact no	int	not null
e-mail	varchar	not null
location	varchar	not null
login_id	int	not null
images	blob	
no of workers	int	not null

Bank details table

FIELD	TYPE	CONSTRAINTS
orphanage_id	int	foreign key
bank	varchar	not null
e-payment no	varchar	not null
contact no	int	not null

Child details table

FIELD	TYPE	CONSTRAINTS
child_id	int	primary key
name	varchar	not null
nick name	varchar	not null
age	varchar	not null
gender	varchar	not null
birth date	varchar	not null
joined date	varchar	not null
known location	varchar	not null
blood group	varchar	not null
orphan type	varchar	not null
adoption type	varchar	not null
images	blob	

Child health report table

FIELD	TYPE	CONSTRAINTS
medical status	varchar	not null
diseases	varchar	not null
disabilities	varchar	not null
height	varchar	not null
weight	varchar	not null
medicines	varchar	not null
child_id	int	foreign key

Child health report table

FIELD	TYPE	CONSTRAINTS
help request id	int	primary key
orphanage_id	int	foreign key
req_type	varchar	not null
data	varchar	not null
req_status	varchar	default : requested

Adoption request table

FIELD	TYPE	CONSTRAINTS
adoption id	int	primary key
sender_id	int	foreign_key
child_id	int	foreign_key
req_status	varchar	default : requested

User registration table

FIELD	TYPE	CONSTRAINTS
Sender_id	int	primary_key
name	varchar	not null
contact no	int	not null
e-mail	varchar	not null
location	varchar	not null
login_id	int	foreign key
images	blob	

SYSTEM TESTING & IMPLEMENTATION

6.1 System testing

System testing is a level of software testing where a complete and integrated software is tested. The purpose of this test is to evaluate the system's compliance with the specified requirements. It makes a logical assumption that if all the parts of the system are correct, the goal will be successfully achieved. The candidate system is subject to a variety of tests: online response, volume, stress, recovery and security and usability test.

It is the process of exercising or evaluating a system by manual or automatic means to verify that it satisfies the specified requirements or to identify the difference between expected and actual results. The testing activities are aimed at convincing the customer through demonstration and actual use that the software is a solution to the original problem and that both the product and the process that created it are of high quality. It is also used to find and eliminate any residual errors from previous stages and the operational reliability of the system.

Preparation of Testing Data

Software testing is a crucial element of software quality assurance and represents the ultimate review of specification, design and coding. Testing represents an interesting anomaly for the software. During earlier definition and development phases, it was attempted to build software from abstract concepts to tangible implementation. The testing responsible for ensuring that the product that has been built performs the way that the detailed design documentation specifies.

Methods of Testing

Testing is the process of finding bugs in a program. It helps to improve the quality of the software. It has to be done thoroughly and with the help of specialist testers. System testing is a process of checking whether the developed system is working according to the objectives and requirements. The used testing methods are:

• Unit Testing

• Integration Testing

• Validation Testing

• Output Testing

• User Acceptance Testing

Unit Testing

The first level of testing is called as unit testing. Here the different modules are tested and the specification produced during design for the modules. Unit testing is essential for verification of the goal and to test the internal logic of the modules. Unit testing is conducted to different modules of the project. Errors were noted down and corrected down immediately and the program clarity was increased. The testing was carried out during the programming stage itself. In this step each module is found to be working satisfactory as regard to be expected out from the module.

Integration Testing

The second level of testing includes integration testing. It is a systematic testing of constructing structure. At the same time tests are conducted to uncover errors with the interface. It need not to be the case, that software whose modules when run individually showing results will also show perfect results when run as a whole. The individual modules are tested again and the results are verified. The goal is to see if the modules integrated between the modules. This testing activity can be considered as testing the design and emphasizes on testing modules interaction.

Validation Testing

The next level of testing is validation testing. Here the entire software is tested. The reference document for this process is the requirement and the goal is to see if the software meets its requirements. The requirement document reflects and determines whether the software functions as the user expected. At culmination of integration testing, software is completely assembled as a package and corrected and a final series of software test validation test begins. The proposed system under construction has been tested by using validation testing and found to be working satisfactory. Data validation checking is done to see whether the corresponding entries made in different tables are done correctly. Proper validation checks are done in case of insertion and updating of tables, in order to see that no duplication of data has occurred. If any such case arises proper warning message will be displayed.

Output Testing

The output of the software should be acceptable to the system user. The output of requirement is defined during the system analysis. Testing of the software system is done against the output and the output testing was completed with success.

User Acceptance Testing

An acceptance test has the objective of selling the user on the validity and reliability of the system. It verifies that the system procedures operate to system specification and the integrity of the vital data is maintained.

6.2 System implementation

System implementation is the final phase 1.e., putting the utility into action. Implementation is the state in the project where theoretical design turned into working system. Implementation involves the conversion of a basic application to complete replacement with a computer system. It is the process of converting to a new or revised system design into an operational one. During the design phase, the products structure, its undergoing data structures, the general algorithms and the interfaces and control/data linkages needed to support communication among the various sub structures were established.

Implementation process is simply a translation of the design abstraction into the physical realization, using the language of the target architecture.

Implementation includes all those activities that take place to convert from the old system to the new. The new system may be totally new replacing an existing manual or automated system, or it may be major modification to an existing system. In either case, proper implementation is essential to provide are liable system to meet organizational requirements. There are three types of implementation:

• Implementation of a computer system to replace a manual system.

• Implementation of a new computer system to replace an existing one.

• Implementation of a modified application to replace an existing one, using the same computer.

The common approaches for implementation are:

Parallel Conversion

In parallel conversion the existing system and new system operates simultaneously until the project team is confident that the new system is working properly. The outputs from the old system continue to be distributed until the new system has proved satisfactorily parallel conversion is a costly method because of the amount of duplications involved.

Direct Conversion

Under direct conversion method the old system is discontinued altogether and the new system becomes operational immediately. A greater risk is associated with direct conversion is no backup in the in the case of system fails.

Pilot Conversion

A pilot conversion would involve the changing over of the part of the system either in parallel or directly. Use of the variation of the two main methods is possible when part of the system can be treated as a separate entity.

User Training

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. For this purpose user manuals are prepared and handled over to the user to operate the developed system. Thus the users are trained to operate the developed system. In order to put new application system into use, the following activities were taken care of:

• Preparation of user and system documentation.

• Conducting user training with demo and hands on.

• Test run for some period to ensure smooth switching over the system.

The major implementation procedures are:

• Test plans

• Training

• Conversion

Test Plans

The implementation of a computer based system requires that the test data be prepared and the system and its elements be tested in a structured manner.

Training

The purpose of training is to ensure that all the personal who are to be associated with the computer based business system possesses the necessary knowledge skills. As the system provides user friendliness only basic training is needed.

Conversion

It is the process of performing all of the operations that results directly in the turnover of the new system to the user. Conversion has two parts:
The creation of a conversion plan at the start of the development phase and the implementation of the plan throughout the development phase. The creation of a system change over plan at the end of the development phase and the implementation of the plan at the beginning of the operation phase.

CODING

When large volumes of data are being handled, it is important that items must be identified, stored or selected easily and quickly. To accomplish this each data item must have a unique identification and must be related to other item of data of the same type. Code can provide briefs identification of data item and replace longer description that would be more specific and manipulated. A code has a group of characters used to identify an item of data

CODE PLANS

The code plans identifies the particular characteristic that need to be continued with in the code. Whatever be the nature of code they must be:

• Expandable
This code must provide space for additional entries that may be required.

• Precise
This code identifies a specific form.

• Concise
This code must be brief and yet adequately describe this item.

• Convenient
This code must be easy to encode and decode.

• Meaningful
This code must be useful to the people dealing with it.

SYSTEM MAINTAINANCE

7.1 Corrective Maintenance

Corrective maintenance refers to changes made to repair defects in the design, coding, or implementation of the system. Corrective maintenance is often needed for repairing processing or performance failures or making change because of previously uncorrected problems or false assumptions. Most corrective maintenance problems surface soon after the installation. When corrective maintenance problems surface, they are typically urgent and need to be resolve to curtail possible interruptions in normal business activities.

7.2 Adaptive Maintenance

Adaptive maintenance involves making changes to an information system to evolve its functionality or to migrate it to different operating environment. Adaptive Maintenance is usually less urgent than corrective maintenance because of business and technical changes typically occur some period of time.

7.3 Perfective Maintenance

Perfective maintenance involves making enhancements to improve processing performance, interface usability, or to add desired, but not necessarily required, system features. Many system professionals feel that perfective maintenance is not really the maintenance but new development.

7.4 Preventive Maintenance

Preventive maintenance is the only maintenance activity which is carried out without formal maintenance request from the user. When a software company or maintenance agency realizes that the methodologies used in a program have become obsolete, it may decide to develop or modify parts of the program, which do not confirm to the current trends. Of these types, more time and money is spending on perfective than on corrective and adaptive maintenance together.

FUTURE ENHANCEMENT

8. FUTURE ENHANCEMENT

The future vision for the Campus Recruitment App is both ambitious and forward-thinking, aiming to revolutionize the landscape of recruitment technology. Here are the key features and technologies that will drive its evolution:

1. AI and ML-driven Matching:

- The incorporation of advanced AI and ML algorithms will revolutionize the matching process. These technologies will analyze vast datasets to provide more accurate and personalized recommendations, ensuring a better fit between student profiles and job opportunities.

2. Predictive Analytics for Career Planning:

- Predictive analytics will be harnessed to anticipate job market trends. This proactive approach will empower students to align their skills with emerging industry demands, fostering a generation of graduates well-prepared for the evolving employment landscape.

3. Immersive VR and AR Experiences:

- Virtual and augmented reality features will elevate the recruitment experience. Immersive virtual career fairs and interactive company presentations will engage both students and recruiters, creating a more dynamic and memorable platform for talent acquisition.

4. Diversity and Bias Mitigation Features:

- To promote diversity and eliminate biases, the app will introduce innovative features. These may include anonymized application processes, diversity benchmarks, and AI tools designed to ensure fair and equal opportunities for candidates from all backgrounds.

5. Blockchain Credential Verification:

- The exploration of blockchain technology for academic and professional credential verification will bring unparalleled security and transparency to the hiring process. This feature will instill trust by providing an immutable and verifiable record of candidates' qualifications.

6. Continuous Feedback and Sentiment Analysis:

- The integration of continuous feedback mechanisms and sentiment analysis tools will create a user-centric design approach. By gathering insights from users, the app will adapt and refine its features, ensuring an evolving and responsive platform that meets the changing needs of its user base.

7. Personalized Learning Paths:

- The app will offer personalized learning paths to students based on their career goals and industry trends. Through AI-driven recommendations, students will receive guidance on additional courses, certifications, or skills development to enhance their employability.

As the Campus Recruitment App evolves with these cutting-edge enhancements, it aspires to be a trailblazer in the recruitment technology landscape. Beyond connecting talent with opportunities, the app aims to shape the future of recruitment by embracing innovation, fostering inclusivity, and providing a user experience that sets new standards in the dynamic intersection of education and employment.

CONCLUSION

9. Conclusion

In conclusion, CareConnect stands as a beacon of hope and compassion in addressing the needs of orphanages and orphaned children. By leveraging technology, the app provides a seamless platform for communication, support, and connection between those in need and those willing to make a positive impact. With its dual modules catering to both orphanages and users, CareConnect facilitates the fulfillment of vital needs such as food, clothing, education, and fundraising, while also offering opportunities for long-term support through initiatives like adoption.

Through its innovative approach, CareConnect not only bridges the gap between orphanages and caring individuals but also cultivates a sense of community and empowerment, ultimately working towards a brighter future for orphaned children worldwide. As we move forward, the potential for CareConnect to continue expanding its reach and impact is immense. By fostering partnerships, increasing user engagement, and continuously refining its features based on user feedback, CareConnect can further enhance its ability to serve orphanages and orphaned children effectively. With its commitment to compassion and connection at its core, CareConnect has the potential to transform the lives of countless orphaned children, offering them hope, support, and the opportunity for a better tomorrow.

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10.BIBLIOGRAPHY

Books/Article

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->>Marco. L. Napoli :Beginning Flutter

->>Thomas Bailey: Flutter for Beginners

Web Resources

->> https://www.flutter.dev/

->> https://github.com/

->> https://Codelabs.developers.com/

->> https://codelabs.developers.google.com

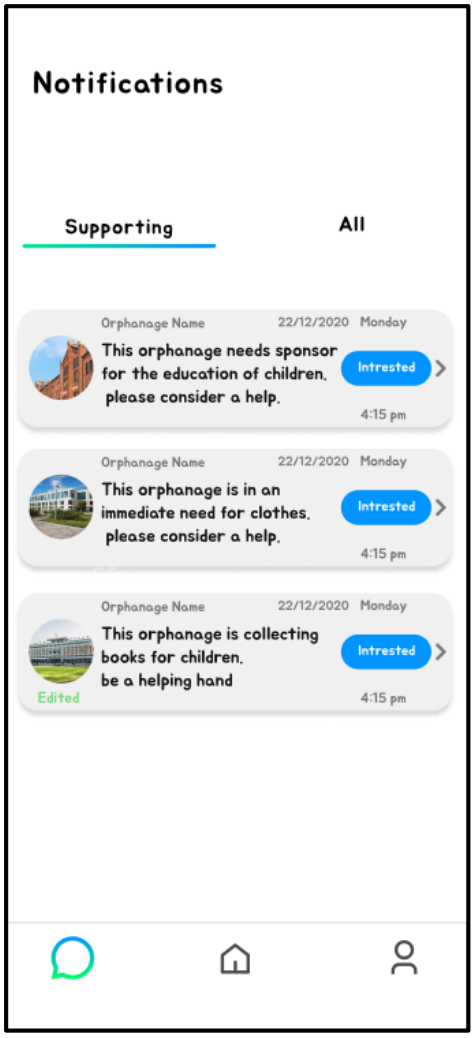
->> https://firebase.google.com/docs

->> https://Codelabs.developers.com/

->> https://dart.dev/guides/language/

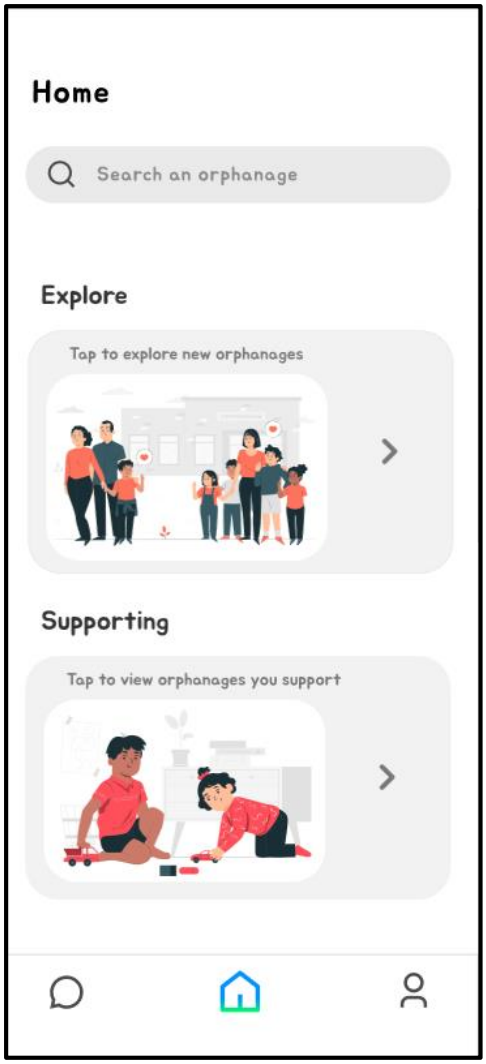
APPENDIX

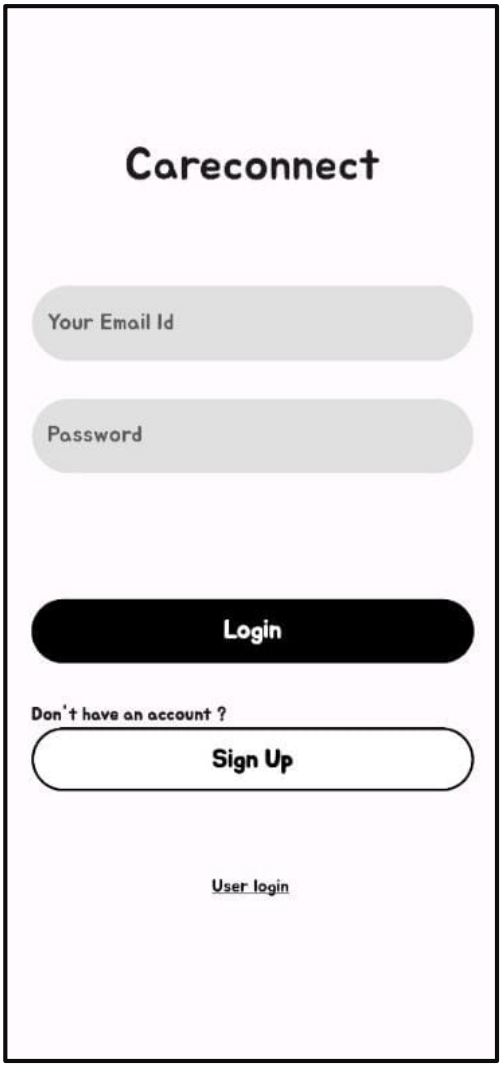
11.1 Sample Screenshots



Notifications page

User profile page





User home page

Orphanage login page

11.2 Sample Codes

Home Page

```
import package:blogui/post.dart';

import package:flutter/material.dart";

class Home extends Stateful Widget {

const Home((Key? key)): super(key: key);

@override

State<Home> createState() => _HomeState();

}

@override

Widget build(BuildContext context) {

return Scaffold(

body: Column(

mainAxisSize: MainAxisSize.max,


children: [

SizedBox(

height: 40,

),

Container(


padding: EdgeInsets.symmetric(horizontal: 20),


child: Row(

mainAxisAlignment: MainAxisAlignment spaceBetween,

children: [
```

```
Text("Blog App",style: TextStyle(
fontSize: 28,
fontWeight: Font Weight.w500,
),
),
IconButton(onPressed: () (), icon: Icon(Icons.account_circle,size: 32,).),
IconButton(onPressed: () {}, icon: Icon(Icons.add,size: 50,).),
],
),
),
Expanded(child: ListView.builder(
itemCount: 5,
itemBuilder: (context,index) {
return Padding(
padding: const EdgeInsets.symmetric(horizontal: 10).
child: Post(),
),
},
),
),
],
),
floating ActionButton: Floating ActionButton.extended(onPressed: 0){
},label: Row(
children: [
```

```
Icon(Icons.upload,size: 25,

),

SizedBox(width: 5,),

Text("Create Blog",style: TextStyle(fontSize: 14),

),

],


),


),

),

}

}
```